

ginning. The Indian defence players would require considerable handholding from the Indian government to help them become a major global player like China's NORINCO, South Korea's Korean Aerospace, and Turkey's Turkish Aerospace Industries, to achieve a degree of self-reliance.

India is hedging its bets to make its military battle-ready for future warfare. Air Marshal Choudhury sees both manned and unmanned systems working in tandem. "When the risk is high for crewed platforms, it will be offset by missiles and drones. It will be a mixed strategy (in the future). Statements like 'contact war is over' are overenthusiastic comments given without adequate thought."

However, in the absence of domestic manufacturing capability for high altitude long endurance drones, India has been forced to buy 31 MQ-9B SkyGuardians/SeaGuardians under a \$3.9 billion deal with the US. With HAL already tied up with the production of LCA fighter jets—and facing delays due to late aero-engine deliveries from GE Aerospace—the Indian government is looking at private players for the fifth-generation Advanced Medium Combat Aircraft (AMCA) programme, estimated to cost around \$1.58 billion. Those in the fray include the Tata Group, a consortium led by Bharat Forge Ltd, and another led by Larsen & Toubro.

## DRONE MANUFACTURING IN INDIA

Ukraine made drones a battlefield weapon, but after losing about 10,000 drones a month, it scaled up its drone capability with external help. The Indian military drone industry is only getting started. The Indian military's shift towards greater automation and unmanned systems was pushed by Operation Sindoor against Pakistan in 2025.

India has a large industrial base for civilian drones, but only three to five manufacturers have the capacity to produce military-grade drones. At present, 80% of drones manufactured in India are for ISR (intelligence, surveillance, reconnaissance), 15% are attack drones, and the remaining 5% are logistics drones.

Sarjan Shah, Managing Director of Shield AI, considers manned-unmanned teaming to be the future of airpower. In 2024, Shield AI, a US drone maker, teamed with JSW Defence to manufacture V-BAT drones in India. It is investing \$90 million to establish a production facility in Hyderabad. The company offers Vertical Take-Off and Landing autonomous aircraft, which removes dependence on long, fixed runways, ensuring survivability in the modern battle-space.



**"Drone technology and software algorithms will need to go through rigorous testing to reach that maturity of robust, consistent decision making"**

**DIPESH GUPTA**  
MANAGING DIRECTOR,  
VEDA AERONAUTICS



**"Not every component must be made in India from day one, but critical systems must be auditable, secure, replaceable, and free of foreign dependencies"**

**SARJAN SHAH**  
MANAGING DIRECTOR,  
SHIELD AI

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The other challenge facing the Indian drone industry is the supply chain gaps. The recent West Asia conflict delayed shipments of Israeli-made drone components. The thermal cameras and infrared optics are dependent on germanium lenses, a semiconductor controlled by China. Recently, Hyderabad-based start-up Aeon Spacelabs developed the country's first germanium-free lens, Lumira E40I50, for ISR missions. "Not every component must be made in India from day one—but every critical system must be auditable, secure, replaceable, and free of opaque foreign dependencies," says Shah.

For India, maximising deterrence with finite defence capital may not lie in choosing between fighter jets and drones, but in investing in a mix that offers the greatest strategic return per rupee. **BT**